

HALO: AN ULTRA LOW COST GRADIENT AMPLIFIER FOR LOW FIELD MRI

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INTRODUCTION

- Halbach low-field scanners lower cost, but accessible systems also require affordable, reproducible electronics.
- Commercial gradient amplifiers exceed the power and cost needs of small, passively cooled coils.
- Open-source designs (e.g., OSII [1]) target higher currents and add monitoring and protection stages, increasing cost and complexity.
- **Aim:** a gradient amplifier under USD 100/channel that is simple, reproducible, locally manufacturable, and suited to low-field, small-FOV MRI.

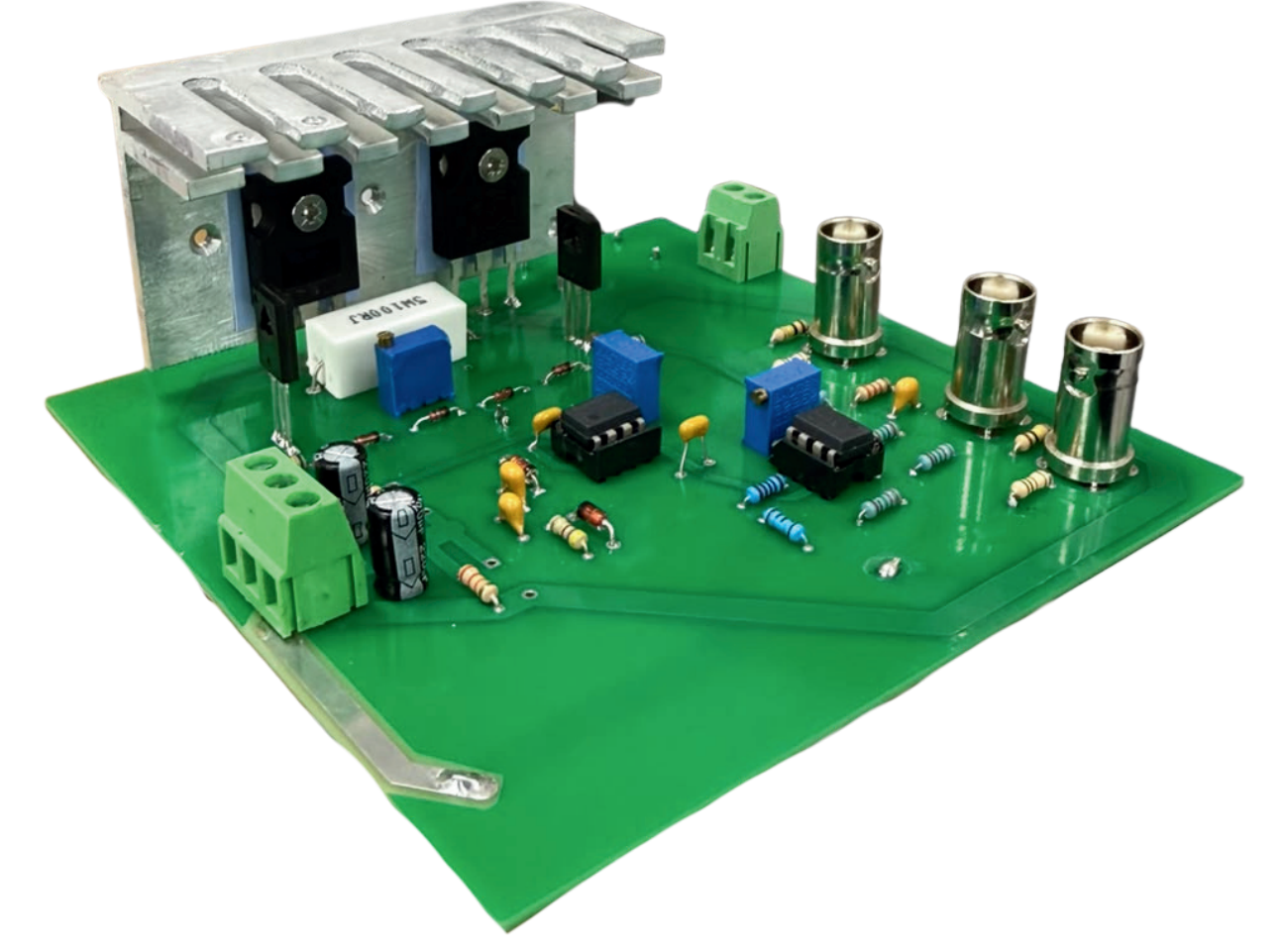


Figure 1: Single-channel amplifier circuit

METHODS

Design requirements and architecture

- Load: Gradient coils (0.2 Ω, 10 μH, 5 mT/(m·A)) driven with trapezoids up to ±6 A and 0.5 ms rise/fall times.
- Supply: ±12 V from two 12 V, 7 Ah batteries, avoiding ground noise, switching EMI and ripple.
- Controller: Closed-loop PI (UA741CP op-amp) following Evetts and Conradi [2], ensuring zero steady-state error on the RL load.
- Power stage: Class AB adapted from Self [3], providing symmetric bidirectional current, low distortion and low output impedance (Figs. 2–3).

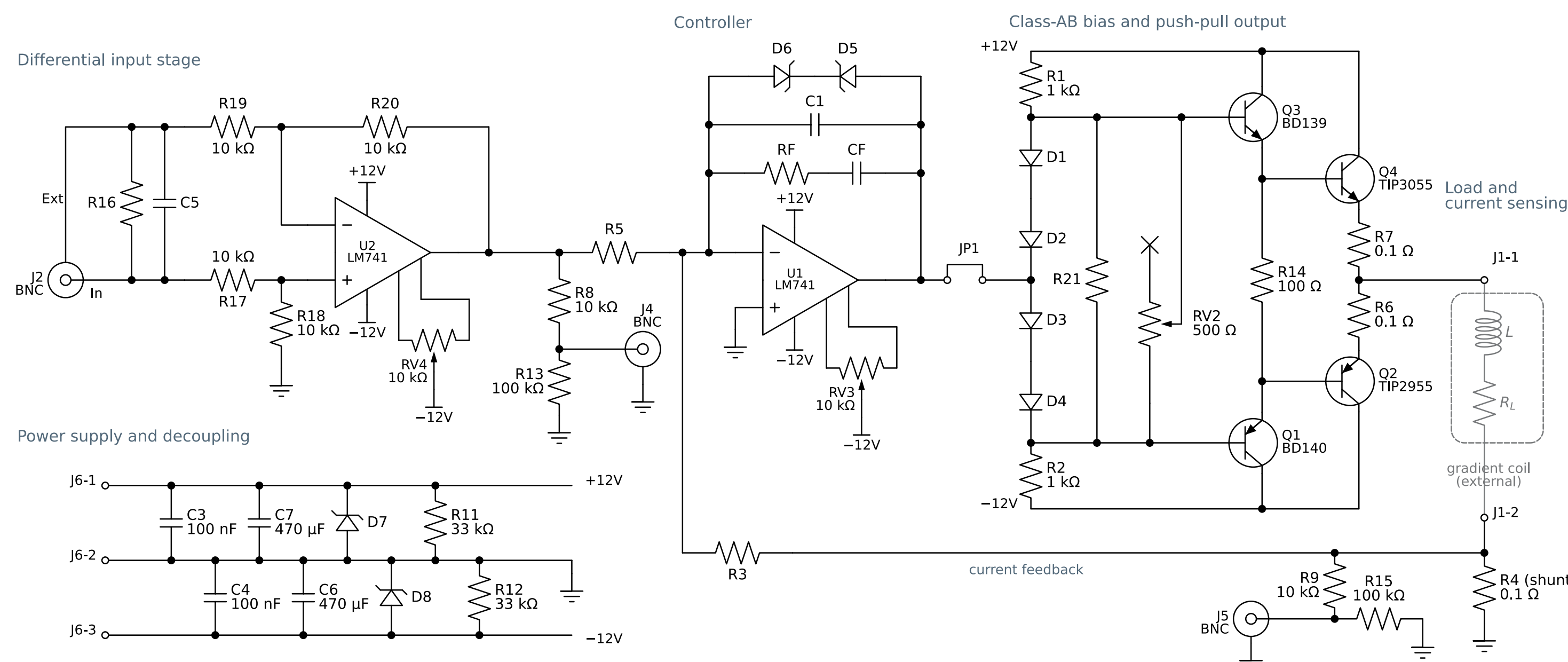


Figure 2: Implemented circuit schematic



Figure 3: Gradient coil used as circuit load and the closed-loop control equations

$$C(s) = -\frac{1}{10R_3} \cdot \frac{sC_f R_f + 1}{s(sC_f R_f C_1 + (C_1 + C_f))} \quad (1)$$

$$T_0(s) = \frac{-\frac{1}{10R_3} \cdot \frac{sC_f R_f + 1}{s(sC_f R_f C_1 + (C_1 + C_f))} \cdot \frac{1}{R_L + sL}}{1 + \frac{1}{10R_3} \cdot \frac{sC_f R_f + 1}{s(sC_f R_f C_1 + (C_1 + C_f))} \cdot \frac{1}{R_L + sL}} \quad (2)$$

$$T_0(s) = \frac{-1}{1 + 10R_3 R_L \cdot s(sC_f R_f C_1 + (C_1 + C_f))} \quad (3)$$

Power stage

- Current boost: Darlington class AB stage amplifying the op-amp's ~20 mA output.
- Output transistors: TIP3055/TIP2955 (15 A, 90 W) driven by a BD139/BD140 pair.
- Bias: Four 1N4148 diodes, adjusted with RV2.

Controller tuning

- Transfer functions: Controller Eq. (1) yielding closed loop Eq. (2), reduced to Eq. (3) when $C_f R_f = L/R_L$.
- Bandwidth: At least 10 kHz, targeted for tracking 0.5 ms ramps.
- Closed-loop design tool on github repository (see QR code).

REFERENCES

- [1] D. H. de Gans, *Gradient Amplifier Design, Open Source Imaging Initiative design document*, 2020.
[2] N. Evetts and M. S. Conradi, "Low-cost gradient amplifiers for small MRI systems," *Journal of Magnetic Resonance*, vol. 335, 107127, 2022.
[3] D. Self, *Audio Power Amplifier Design*, 6th ed., Focal Press, 2013.

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RESULTS

- Construction: Built from local through-hole parts on a simplified PCB for about USD 70/channel (about USD 210 for 3 axes).

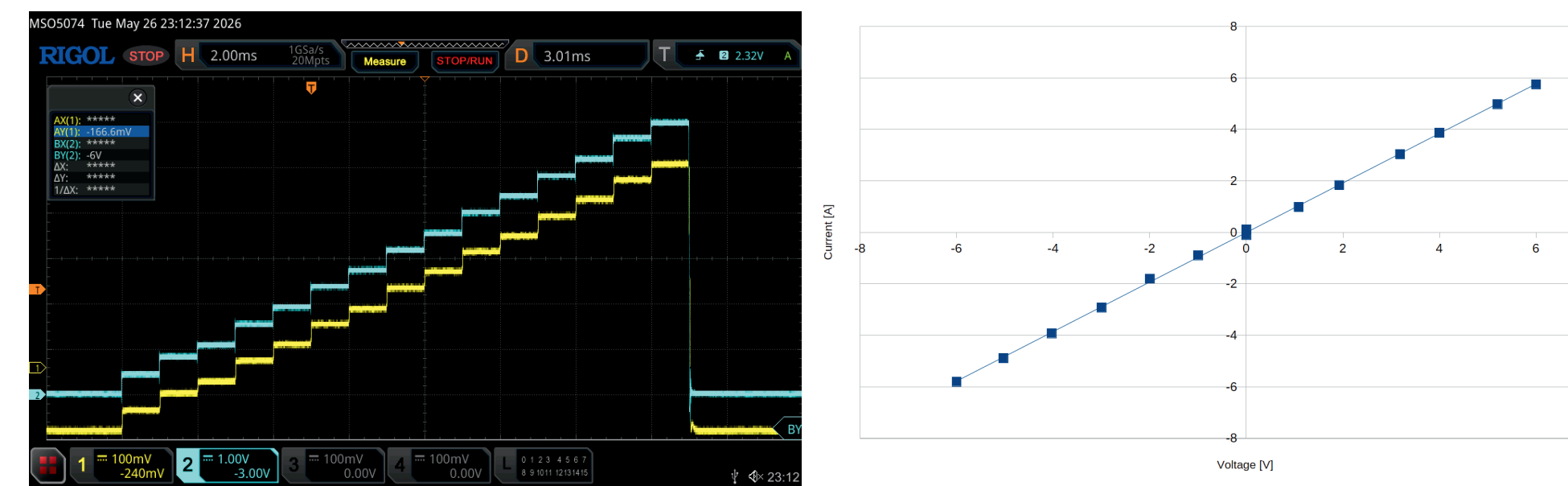


Figure 4: Stepped current response (Cyan: voltage input; yellow: measured output current) and current-voltage linear fit

- Rise time: Measured at 0.06 ms for a 6 A step (30 mT/m) against a 0.5 ms target (Fig. 5), enabling faster switching, shorter readouts and less off-resonance.

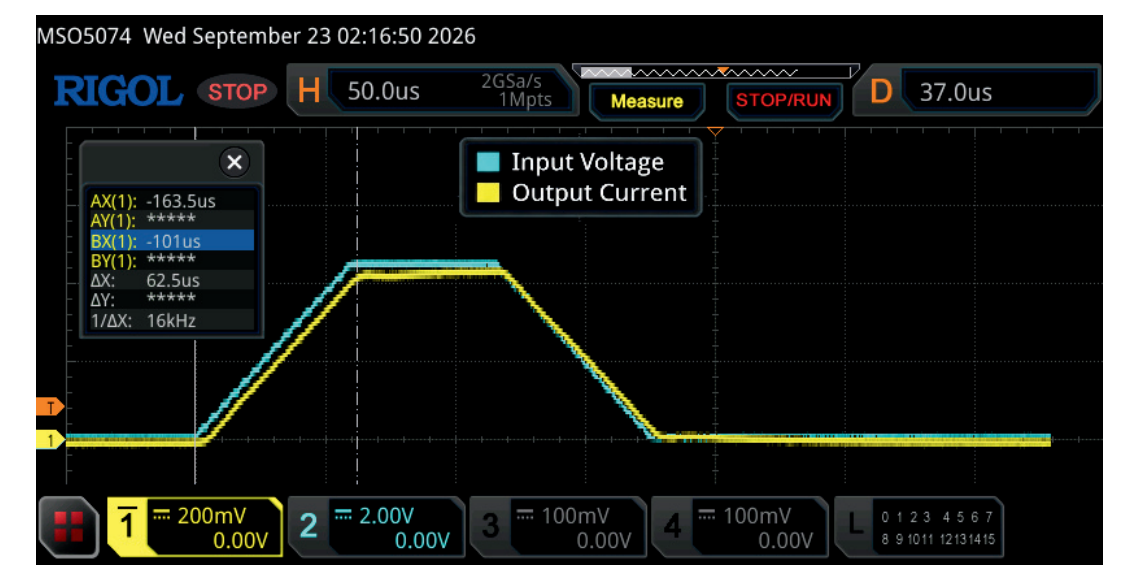


Figure 5: Circuit response to a 6 A step with 0.06 ms rise time.

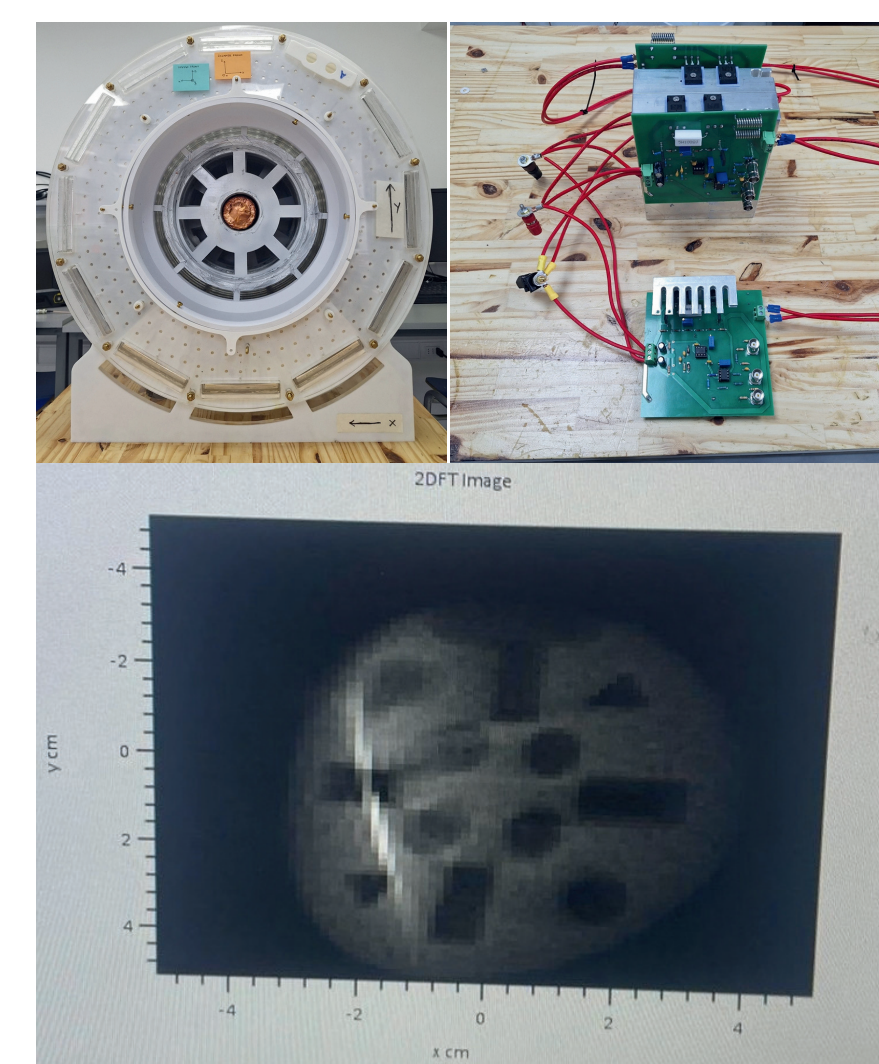


Figure 6: 46 mT Halbach setup and oil phantom results

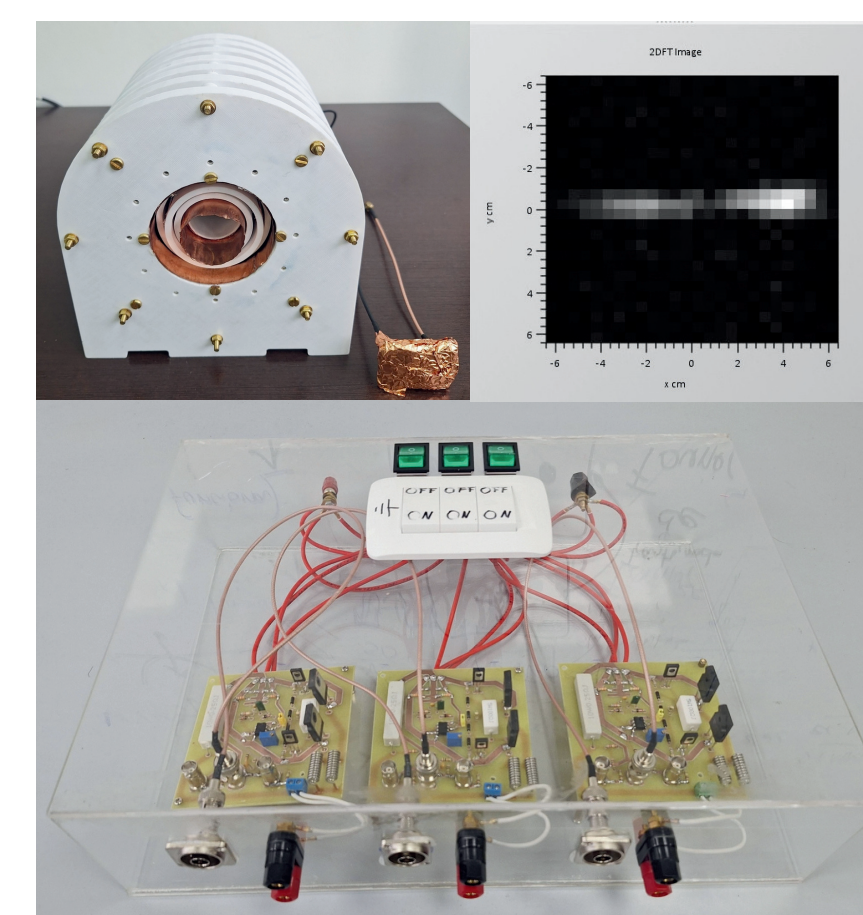


Figure 7: 50 mT ERNIE setup and two oil tubes results

- **MRI integration:** Tested with three amplifiers on a shared ±12 V battery pack driving the x/y/z coils of an in-house 46 mT Halbach scanner, imaging an oil phantom (Fig. 6).
- **ERNIE:** Integrated into the CAMERA-IMAGINE educational toolkit (piloted at the ESMRMB 2025 Workshop), yielding gradient-encoded phantom images (Fig. 7) and showing readiness for open educational platforms.
- **Limitation:** Tuning of the PI controller required for each load and target bandwidth; a design tool for choosing the parameters is available on GitHub, together with the rest of the project files (see QR code).

CONCLUSIONS

- Costs under USD 100/channel for low-field MRI.
- Provides linear current control, delivering ±6 A at 0.1 A/μs (500 T/m/s) into 0.2 Ω, 10 μH.
- Integrates into a 46 mT Halbach scanner and into ERNIE.
- Offers a simple, reproducible route to accessible gradient hardware.

